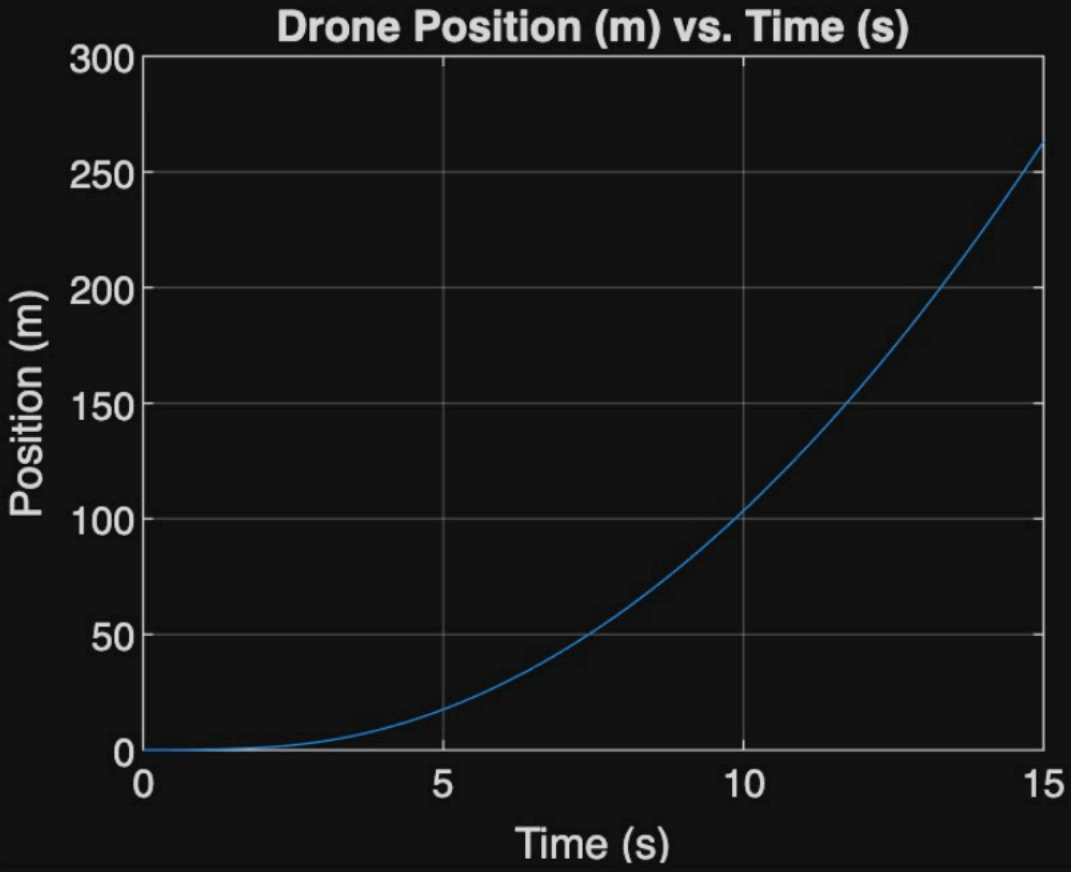
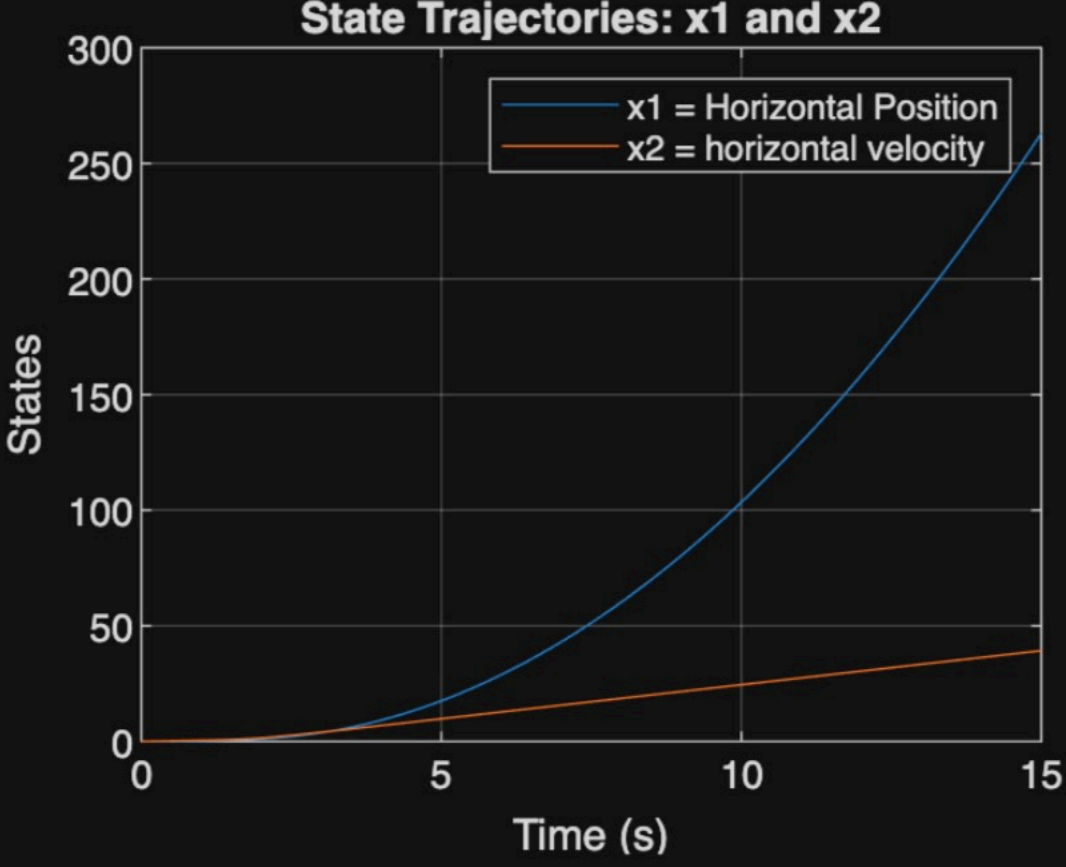
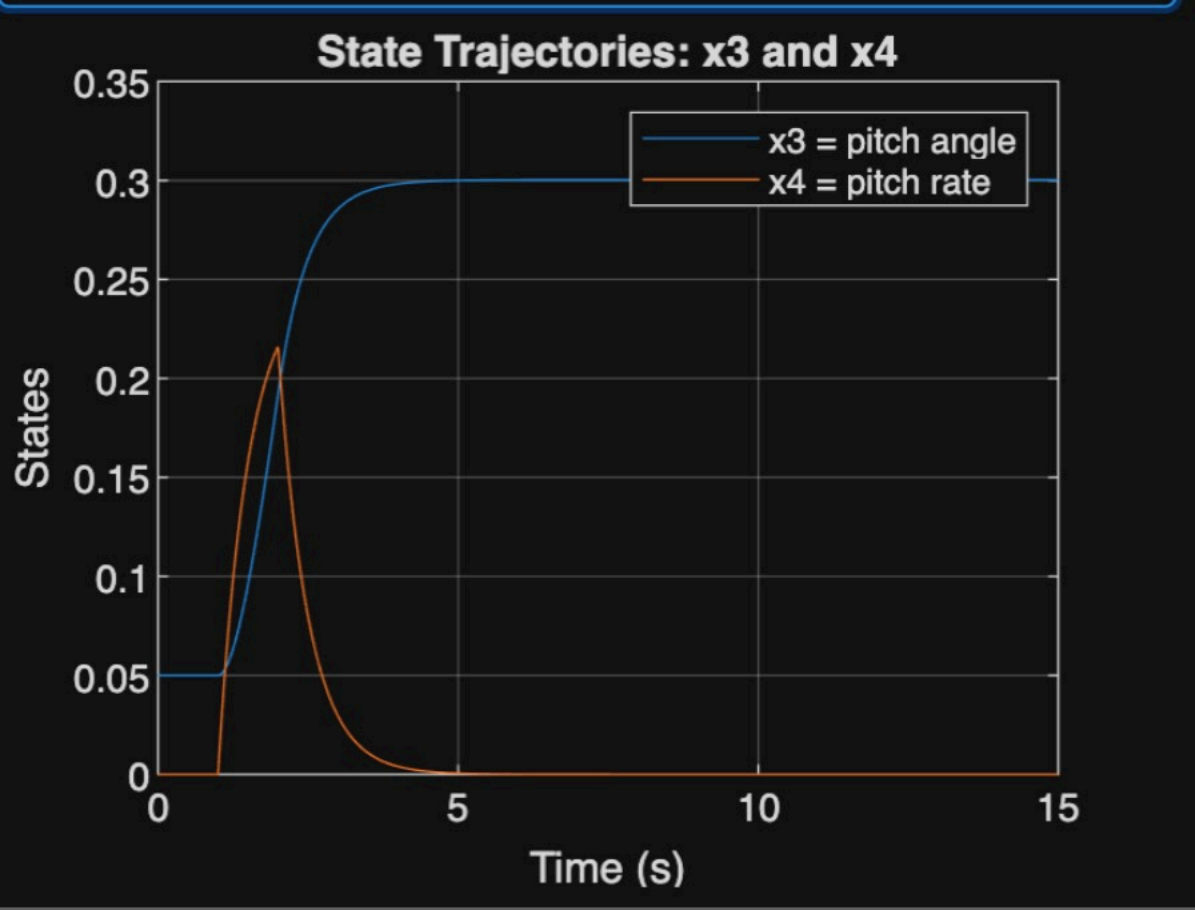

Drone Position (m) vs. Time (s)
300
250
200
150
100
50
0
Position (m)
0
5
10
15
Time (s)


State Trajectories: x1 and x2
x1 = Horizontal Position
x2 = horizontal velocity
States
Time (s)
300
250
200
150
100
50
0
0
5
10
15


State Trajectories: x3 and x4
x3 = pitch angle
x4 = pitch rate
States
Time (s)
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
5
10
15

The above results make sense. x2 is rightly the derivative and rate of change of x1. x4 has a brief jump as u (input torque) is applied, which results in x3 being affected. As the torque input is released, x3 and x4 both settle.